Project Explanation, Business Impact and Conclusion

1. Project Overview

EcoDriveAI+ is an AI-driven electric vehicle energy optimization system designed to understand how driving behaviour, vehicle conditions, traffic, and environmental factors influence EV energy consumption.

The objective of the project is not only to predict energy consumption, but also to transform vehicle telemetry into actionable recommendations for more efficient driving.

The system follows a complete data-to-decision pipeline:

Vehicle Data → Data Cleaning → Feature Engineering → Machine Learning → Driving Behaviour Analysis → Energy Prediction → EcoDrive Score → Recommendations → Energy-Saving Simulation → Dashboard

The project combines predictive machine learning, unsupervised learning, behavioural analytics, counterfactual analysis, and interactive visualization.

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2. What Was Implemented in the Code?

2.1 Data Loading and Data Quality Analysis

The project begins by loading the EV energy-consumption dataset and Tesla telemetry data.

The data was inspected for:

- Missing values
- Duplicate records
- Incorrect data types
- Numerical and categorical variables
- Battery measurements
- Speed and acceleration information
- Environmental conditions
- Vehicle characteristics
- Driving-related measurements

This step was important because machine-learning models are only reliable when the input data is properly prepared.

2.2 Data Cleaning and Standardization

The datasets were cleaned and standardized before modelling.

The code handles:

- Missing values
- Invalid numerical values
- Duplicate observations
- Inconsistent column names
- Numerical conversion
- Categorical variables
- Missing telemetry variables

This makes the modelling pipeline more robust and allows different sources of vehicle data to be processed in a consistent format.

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3. Feature Engineering

One of the main contributions of this project is the creation of additional features from raw vehicle telemetry.

Instead of relying only on raw variables such as speed and acceleration, the system creates behavioural and driving-stress indicators.

Examples include:

Acceleration behaviour

The system identifies:

- Harsh acceleration
- Harsh braking
- Normal acceleration
- Acceleration magnitude

These features help identify inefficient driving patterns.

Jerk

Jerk represents how quickly acceleration changes.

A high jerk value can indicate sudden changes in driving behaviour.

The project therefore calculates:

Jerk = Change in acceleration over time

and also calculates absolute jerk to measure the magnitude of these changes.

Speed variability

Rolling statistics are used to measure how unstable the vehicle speed is over a short period.

This helps distinguish smoother driving from highly variable driving.

Traffic stress

The system combines speed changes and acceleration-related information to create a traffic-stress indicator.

This allows the model to consider the effect of stop-and-go conditions rather than treating every trip as if it occurred under identical traffic conditions.

Aggressive Driving Score

A behavioural score is created from acceleration, braking, jerk and speed-related features.

This provides a numerical representation of driving aggressiveness.

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4. Energy Consumption Analysis

The project analyses EV energy consumption and energy intensity.

Energy intensity is particularly important because total energy consumption alone does not tell us whether a vehicle was efficient.

For example:

- A vehicle travelling a longer distance naturally consumes more energy.
- A shorter trip with extremely high energy consumption may indicate inefficient driving.

Therefore, the project also considers energy consumption relative to distance.

A useful business metric is:

Energy Intensity = Energy Consumption / Distance Travelled

This allows trips with different distances to be compared more fairly.

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5. Exploratory Data Analysis

The project performs exploratory analysis to understand relationships between:

- Speed
- Acceleration
- Battery state of charge

- Battery voltage
- Temperature
- Traffic
- Weather
- Vehicle characteristics
- Distance
- Energy consumption

Correlation analysis and visualizations were used to identify potentially important relationships before machine learning.

This stage provides interpretability and helps ensure that the modelling process is based on meaningful vehicle and driving characteristics.

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6. Machine Learning Models

Multiple machine-learning models were implemented rather than relying on a single algorithm.

The models include:

- Dummy Regressor
- Linear Regression
- Random Forest
- HistGradientBoosting
- XGBoost

This allows the models to be compared objectively.

The evaluation metrics include:

MAE — Mean Absolute Error

Measures the average magnitude of prediction errors.

RMSE — Root Mean Squared Error

Penalizes larger prediction errors more strongly.

R^2 — Coefficient of Determination

Measures how much variation in energy consumption is explained by the model.

The final model is selected based on validation performance rather than simply assuming that a more complex model will always be better.

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7. Model Evaluation

The model is evaluated using unseen validation/test data.

The project compares:

- Actual energy consumption
- Predicted energy consumption
- MAE
- RMSE
- R^2

An actual-versus-predicted visualization is also generated.

Residual analysis is used to identify whether the model systematically underpredicts or overpredicts energy consumption.

This is important because a high R^2 alone does not guarantee that a model is reliable.

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8. Feature Importance and Explainable AI

The project also analyses which variables contribute most strongly to the predictions.

Feature importance is calculated for tree-based models, and SHAP analysis is used where applicable.

This moves the project beyond:

"The model predicts energy consumption."

towards:

"The model predicts energy consumption and provides evidence about which driving, vehicle, and environmental factors influence the prediction."

This is particularly important for business applications because decision-makers need explanations rather than unexplained AI predictions.

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9. Tesla Telemetry Analysis

The project applies the developed feature-engineering and prediction pipeline to Tesla telemetry data.

The Tesla data contains real vehicle telemetry measurements such as:

- Speed

- Acceleration
- Battery information
- Odometer information
- Temperature-related measurements
- Cumulative battery discharge

The objective is to demonstrate how the model can be used with real-world vehicle telemetry rather than only a clean modelling dataset.

However, Tesla predictions should be interpreted as **model-based estimates** because the Tesla telemetry does not provide the same directly labelled energy-consumption target used for training.

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10. Driving Behaviour Clustering

Unsupervised machine learning is used to identify different driving behaviours.

K-Means clustering is applied using features such as:

- Speed
- Acceleration
- Jerk
- Speed variability
- Acceleration variability
- Harsh acceleration
- Harsh braking
- Eco-driving indicators

The optimal number of clusters is evaluated using silhouette analysis.

Instead of manually assigning arbitrary cluster labels, the clusters are profiled using their actual behavioural characteristics.

The resulting groups can then be interpreted as behaviours such as:

- Eco
- Normal
- Aggressive

This allows the system to answer:

"How is the vehicle being driven?"

rather than only:

"How much energy is being consumed?"

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11. EcoDrive Score

A key output of EcoDriveAI+ is the EcoDrive Score.

The score converts several driving-behaviour indicators into a simple scale from:

0 = highly inefficient driving

to

100 = highly efficient/smooth driving

The score considers factors such as:

- Harsh acceleration
- Harsh braking
- Jerk
- Speed instability

This makes complex telemetry easier to understand for drivers, fleet managers and business users.

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12. Counterfactual Eco-Driving Simulation

One of the most important analytical components is the counterfactual simulation.

The system creates an alternative scenario in which aggressive driving characteristics are reduced.

For example, the simulation reduces acceleration and jerk intensity and removes the aggressive-driving contribution.

The model then predicts energy consumption under this hypothetical eco-driving scenario.

The difference between:

Predicted current energy consumption

and

Predicted eco-driving energy consumption

is used to estimate potential energy savings.

This allows the system to answer:

"How much energy could potentially be saved if the driver adopted smoother driving behaviour?"

These are **model-based estimated savings**, not measured real-world savings from an intervention.

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13. Recommendation Engine

The project converts analytical results into actionable recommendations.

For example, if the system detects:

- High harsh-acceleration frequency → recommend smoother acceleration.
- High harsh-braking frequency → recommend earlier and smoother braking.
- High jerk → recommend avoiding sudden throttle changes.
- High traffic stress → recommend smoother speed management.
- Low EcoDrive Score → provide additional eco-driving guidance.

This creates a transition from:

Data → Prediction → Insight → Action

which is much more useful in a real business environment than a prediction model alone.

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14. Route Energy Analysis

The project also introduces energy-based route comparison.

Instead of evaluating routes only using:

- Distance
- Travel time

the system considers:

Estimated energy consumption

as an additional optimization objective.

This is important for EVs because the shortest route is not necessarily the most energy-efficient route.

For example, a slightly longer route with:

- smoother traffic,
- fewer stops,
- less aggressive acceleration,
- and better driving conditions

could potentially consume less energy.

The current route scenarios in this project demonstrate the concept. A production implementation would connect this component to real navigation, traffic, elevation, weather and road-condition data.

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15. Interactive Dashboard

A Dash and Plotly dashboard was developed to make the analytical results accessible to non-technical users.

The dashboard provides:

- Model performance information
- Energy-consumption predictions
- Driving-behaviour analysis
- EcoDrive Score distribution
- Traffic analysis
- Energy-related visualizations
- Behaviour filtering

This transforms the project from a machine-learning notebook into a prototype decision-support application.

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16. Business Impact

EcoDriveAI+ can provide value to several business areas.

16.1 Fleet Management

Companies operating EV fleets could use the system to monitor driver behaviour and identify inefficient driving patterns.

Fleet managers could compare:

- Drivers
- Vehicles
- Routes
- Energy intensity
- EcoDrive Scores

This could support data-driven driver coaching and fleet optimization.

16.2 Cost Reduction

Electricity is a major operating cost for EV fleets.

Reducing unnecessary energy consumption could potentially reduce:

- Charging costs
- Energy consumption per kilometre
- Number of charging events
- Operational costs

The financial impact can become significant when the system is deployed across a large fleet.

The IEA also highlights that energy efficiency can provide direct competitive benefits for businesses, particularly where energy costs are significant.

16.3 Better Fleet Utilization

Improved energy efficiency can potentially increase the useful driving range available from the same battery charge.

This can help fleet operators reduce unnecessary charging interruptions and improve vehicle availability.

For delivery, logistics, taxi or mobility companies, even small improvements repeated across thousands of kilometres can become operationally meaningful.

16.4 Driver Performance Management

The EcoDrive Score can provide a measurable way to evaluate driving behaviour.

Instead of simply telling a driver:

"Drive more efficiently."

the company can provide measurable feedback such as:

- EcoDrive Score
- Harsh acceleration rate
- Harsh braking rate
- Energy intensity
- Estimated energy-saving opportunity

This creates a feedback loop between driver behaviour and energy efficiency.

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17. Environmental and Social Impact

The project also has a broader sustainability objective.

EVs already reduce dependence on fossil fuels, but increasing EV adoption also increases electricity demand. The IEA estimates that global EV electricity consumption was around 250 TWh in 2025 and could exceed 1,500 TWh by 2035 under its Current Policies Scenario.

Therefore, improving the efficiency of every kilometre driven becomes increasingly important as the EV fleet grows.

Eco-driving is also recognized as an energy-efficiency measure. The IEA identifies avoiding sudden acceleration, stops and idling as part of eco-driving behaviour.

EcoDriveAI+ contributes to this objective by using data and AI to identify inefficient behaviour and provide personalized recommendations.

The potential societal benefits include:

- Lower energy waste
- Lower operating costs
- More efficient use of EV batteries
- Improved driving awareness
- Potential reduction in transport-related energy demand
- Support for sustainable mobility

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18. Why This Project Is Different

The project is not simply an EV energy prediction model.

It combines several layers:

Layer 1 — Prediction

Predict EV energy consumption.

Layer 2 — Explanation

Identify which factors influence energy consumption.

Layer 3 — Behaviour Detection

Identify driving styles using unsupervised learning.

Layer 4 — Scoring

Convert driving behaviour into an EcoDrive Score.

Layer 5 — Recommendation

Tell the driver or fleet manager what can be improved.

Layer 6 — Counterfactual Analysis

Estimate how energy consumption could change under improved driving behaviour.

Layer 7 — Decision Support

Compare energy implications of different route scenarios.

Layer 8 — Visualization

Present the results through an interactive dashboard.

Therefore, the project moves from a traditional:

Machine Learning → Prediction

approach to:

Machine Learning → Behaviour → Recommendation → Optimization → Business Decision

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19. Limitations

Although the prototype demonstrates the complete concept, several limitations should be acknowledged.

1. Cross-dataset validation

The main energy-consumption model is trained using one dataset and then applied to Tesla telemetry from another data source.

This means the Tesla predictions represent model-based estimates and should not be interpreted as independently validated ground truth.

2. Estimated energy savings

The counterfactual energy-saving results are simulations produced by the model.

They do not prove that a real driver would achieve exactly the same savings.

3. Route optimization

The current route component demonstrates the methodology using route scenarios.

A production system would require real route alternatives, traffic APIs, road elevation, weather and navigation data.

4. Generalization

A production-grade system should be tested across:

- Different EV models
- Different weather conditions
- Different cities
- Different road types
- Different driver populations

5. Real-world intervention testing

The strongest future validation would involve measuring energy consumption before and after drivers receive EcoDriveAI+ recommendations.

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20. Future Development

The project could be extended into a real commercial product by adding:

Real-time vehicle integration

Connect directly to vehicle telemetry APIs.

Mobile driver application

Provide real-time:

- EcoDrive Score
- Driving alerts
- Energy-saving recommendations

Fleet management platform

Allow companies to monitor entire fleets.

Real-time route optimization

Combine:

- Traffic
- Elevation
- Weather
- Road conditions
- Battery SOC
- Charging locations

to recommend energy-efficient routes.

Personalized AI coaching

Create driver-specific recommendations based on historical behaviour.

Predictive charging

Predict when a vehicle will need charging based on:

- Current battery level
- Driving style
- Route
- Traffic
- Temperature
- Historical consumption

Long-term battery analytics

Study whether aggressive driving patterns are associated with increased battery stress or degradation.

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21. Overall Conclusion

EcoDriveAI+ demonstrates how artificial intelligence can be used to move EV energy management from simple monitoring toward predictive and behavioural optimization.

The project combines machine learning, telemetry analysis, clustering, explainable AI, behavioural scoring, counterfactual simulation and interactive visualization.

The key contribution is the connection between **energy consumption and human driving behaviour**.

Instead of only predicting how much energy an EV will consume, the system attempts to explain:

Why energy consumption changes, how the driver is behaving, what could be improved, and what the potential impact of that improvement could be.

From a business perspective, this creates opportunities for fleet operators to monitor vehicle efficiency, identify inefficient driving patterns, provide targeted driver coaching and potentially reduce energy-related operating costs.

From an environmental perspective, improving energy efficiency becomes increasingly relevant as EV adoption grows and electricity demand from transportation increases. The IEA notes that EV electricity demand is expected to grow substantially as the global EV fleet expands.

The most important future step is real-world validation: deploying the system with actual drivers and measuring whether the recommended behavioural changes produce measurable reductions in energy consumption.

Overall, EcoDriveAI+ demonstrates a complete **AI-to-business-impact pipeline**:

Raw EV Telemetry → Data Engineering → Machine Learning → Behaviour Detection → Explainable Insights → EcoDrive Score → Recommendations → Energy-Saving Simulation → Business Decision Support

This makes the project relevant not only as a machine-learning experiment, but also as a prototype for sustainable mobility, EV fleet management and data-driven energy optimization.